

Question ID: 881ef5f5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Medium

Question

If a is the mean and b is the median of nine consecutive integers, what is the value of $|a - b|$?

Rationale

The correct answer is 0. Any nine consecutive integers can be written as $k, k+1, k+2, k+3, k+4, k+5, k+6, k+7, k+8$. The mean of the integers is their sum divided by 9: $\frac{(k+k+1+k+2+\dots+k+8)}{9} = \frac{(9k+36)}{9}$, which simplifies to $k+4$. So $a = k+4$. Since there is an odd number of integers (nine), the median is the integer in the middle when all the integers are ordered from least to greatest: $k+4$. So $b = k+4$. Therefore, $|a - b| = |(k+4) - (k+4)|$, which is 0.